

The Emergence of Cinematic Time: Modernity, Contingency, the Archive

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Zeno's Paradox: The Emergence of Cinematic Time

In the technical language of filmmaking, the term *real time* refers to the duration of a single shot (assuming the shot is neither fast nor slow motion). If the physical film is not cut and its projection speed equals its shooting speed (usually somewhere between sixteen and twenty-four frames per second), the movement on the screen will unfold in a time that is isomorphic with profilmic time, or what is generally thought to be our everyday lived experience of time—hence the term *real*. The time of the apparatus matches, is married to, the time of the action or the scene. This “real time” is marked by an apparent plenitude. No lack or loss of time is visible to the eye or accessible to the spectator. But this temporal continuity is in fact haunted by absence, by the lost time represented by the division between frames. During the projection of a film, the spectator is sitting in an unperceived darkness for almost 40 percent of the running time. Hence, much of the movement or the time allegedly recorded by the camera is simply not there, lost in the interstices between frames. These interstices, crucial to the representation of movement, must themselves remain unacknowledged. The cinema presents us with a simulacrum of time. Nevertheless, knowledge of the indexicality of the cinematic image sustains a belief that something of time, something of movement or its imprint, or, at the very least, its adequate representation, is there.

Because an adequate representation of motion in time appeared to be a powerful result of cinematic technology, the cinema seemed to address, or even resolve, an ancient philosophical argument. Theoretical and philosophical discourses on the cinema have from a very early point paired the cinema's production of continuous movement from discontinuous images with Zeno's paradoxes. In the 1920s Jean Epstein, under the heading “The trans-

mutation of the discontinuous into the continuous, negated by Zeno, but accomplished by the cinematograph,” wrote: “the cinematograph seems to be a mysterious mechanism intended to assess the false accuracy of Zeno’s famous argument about the arrow, intended for the analysis of the subtle metamorphosis of stasis into mobility, of emptiness into solid, of continuous into discontinuous, a transformation as stupefying as the generation of life from inanimate elements.” In 1907 Henri Bergson directly linked a discussion of the cinematographic illusion of motion and Zeno’s paradoxes, and accused both of the same error—that of attempting to reconstitute movement from static states or instants. Real movement, he claimed, escapes the grasp of both: “The movement slips through the interval, because every attempt to reconstitute change out of states implies the absurd proposition, that movement is made of immobilities.”¹

Zeno was the perfect nemesis for Bergson because, as a member of the Eleatic School of pre-Socratic philosophers, he was fully invested in the denial of movement, change, and plurality. His paradoxes were designed to demonstrate the absurdity of a commonsense belief in the reality of such concepts. The paradox of the arrow to which Epstein refers is an account that undermines our belief in its movement. At any given moment, the arrow occupies a space equal to its volume and simply is where it is. Therefore, at any moment it is at rest. Therefore, it is always at rest and never moves. At each point of its course, it is motionless; hence it is motionless during the entire time it is allegedly moving. A second paradox attempts to prove that it is impossible for a person to reach the far end of a stadium, since before attaining the goal the person must reach the halfway mark and, before that, the halfway mark of the halfway mark, and so on, in perpetuity. Because space is infinitely divisible, it is impossible to cross it with a finite number of steps. A third paradox concerns the famous race between Achilles and the tortoise. The tortoise is given a head start of ten units, but Achilles runs ten times faster. Zeno argued that Achilles would never pass the tortoise because every time Achilles attains the point where the tortoise was, the tortoise has progressed a tenth of the distance Achilles has run. The tortoise will always be ahead.

Epstein gives Zeno credit for understanding that “the analysis of movement yielded a collection of stops,” but faults him for failing to envisage the seemingly impossible—their absurd synthesis in the cinema, which de-

molishes the opposition between the continuous and the discontinuous (“Hardly anyone has realized that the cinematic image carries a warning of something monstrous, that it bears a subtle venom which could corrupt the entire rational order so painstakingly imagined in the destiny of the universe”).² Epstein’s discourse is an avant-gardist celebration of this monstrous cinema. Bergson, on the other hand, rejects any claim the cinema might make to representing the truth of time or movement. Zeno’s fallacy finds its technological embodiment in the cinema—in its spatialization of time, its investment in the reality of instants. According to Bergson, all Zeno’s paradoxes are essentially the same, and that one paradox in its different varieties errs in its misrecognition of the essence of movement. The arrow’s movement is not confronted as such but instead is reduced to its trajectory, a line in space tracing the path of the moving body. This trajectory is, indeed, divisible and, furthermore, infinitely divisible, as Zeno’s paradox of the stadium emphasizes. But the trajectory is only the shadow, the trace of the movement, which is in itself indivisible. Since movement is qualitative rather than quantitative, its divisibility is unthinkable. The arrow is not at rest at any point in its course, for those instants of immobility are only potential, not actual. When the arrow does come to rest, its movement is over, and any new movement departing from that point is a qualitatively different movement. Zeno’s mistake lies in assuming that what is true of the trajectory or line is true of the movement, but the trajectory simply subtends the movement—it does not define it. Movement takes place in the interval, in the transition *between* states, not in their accumulation. This explains, for Bergson, the profound unreality of cinematic “real time.” Movement cannot be reconstituted from immobilities.

Yet how does Bergson deal with the fact of cinema? It becomes emblematic, for him, of our ordinary, everyday, pragmatic knowledge of time. Such knowledge is, of necessity, discontinuous and has what Bergson refers to as a cinematographic quality. But for Bergson it is crucial to move beyond that cinematographic impulse to grasp the true nature of duration and movement. The cinema attempts to reconstitute movement with a series of still photographs, but none of these photographs has anything to do with movement. Their gestures are static, frozen. Bergson admits that in order to produce the illusion of movement there must be real movement somewhere. He locates it in the apparatus, the projector, which moves the film forward. The

movement of the projector is always the same and succeeds in abstracting a kind of general movement from the individual, particular movements recorded by the camera:

The process then consists in extracting from all the movements peculiar to all the figures an impersonal movement abstract and simple, *movement in general*, so to speak: we put this into the apparatus and we reconstitute the individuality of each particular movement by combining this nameless movement with the personal attitudes. Such is the contrivance of the cinematograph. And such is also that of our knowledge . . . Whether we would think becoming, or express it, or even perceive it, we hardly do anything else than set going a kind of cinematograph inside us.

Yet, as Bergson later claims, “the movement slips through the interval.”³ The cinematograph can produce only the illusion of mobility.

Bergson's adamant rejection of the cinema as an adequate representation of time poses problems for Gilles Deleuze, who, in his massive two-volume treatment of the cinema, movement, and time, appeals to Bergson for the guiding framework of his discussion. One way of circumventing the problem is to claim, as Deleuze does, that the cinema Bergson dismissed in *Creative Evolution* was a primitive cinema and that “things are never defined by their primitive state, but by the tendency concealed in this state.” Another approach involves the claim that Bergson actually anticipated the cinema in his groundbreaking discussion of movement and duration in the first chapter of *Matter and Memory* (published in 1896). From this point of view, the cinema emerges as a philosophical machine for the demonstration of duration in its truth—for the presentation of “time in the pure state.”⁴

Bergson's misrecognition of the cinema's true capabilities, according to Deleuze, is linked to his insistence upon locating the cinema's access to real movement in the projector, in a homogeneous mechanical time subject to the “abstract idea of a succession” and “copied from space.” Bergson therefore believed that cinematic movement was reducible to the formula “immobile sections + abstract time.” Deleuze claims, on the other hand, that the movement needs to be thought in relation to the spectator rather than in relation to the apparatus, and that, for the spectator, movement is immedi-

ately given in an “intermediate image.” The spectator does not see the succession of photograms but, instead, an intermediate image, which is a “mobile section” not an immobility.⁵ This mobile section is not the illusion of movement but its reality; it is imbued with qualitative change and duration. In Deleuze’s view, the cinema constitutes a massive refutation of Zeno; it does not participate in his error, as Bergson, who ironically produced a philosophy consonant with the cinema, claimed.

Zeno’s paradoxes have provoked extensive and long-lived discussion in mathematics and philosophy. In general, the attempt has *not* been to ascertain the truth or falsity of Zeno’s contention (that there is no such thing as movement or change), but to determine why he was wrong, what fatal flaw marred his reasoning in the paradoxes. Bergson believed that our intuition that movement and change exist is correct and simply requires philosophical corroboration. His entire philosophy is in fact a celebration of movement, change, duration. It should not be surprising that an obsession with the limits of Zeno’s paradoxes should be resuscitated in modernity, or in the wake of the emergence of the cinema. For in modernity time seems to change its contours, to become more insistently present as a problem, to be subject to acceleration and manipulation. Representational systems (art, literature) address themselves to the ephemeral, the contingent, the moment. The credibility of any static universal or eternal is diminished. And, in this context, the cinema seems to offer a direct answer to Zeno in insisting that movement can indeed be born from immobilities. Cinema works by obliterating the photogram, annihilating that which is static. It appears to extract a magical continuity from what is acknowledged to be discontinuous. The moment of fascination in the early cinema is the moment when the still image projected on the screen bursts into movement.⁶

Epstein, writing in the 1920s, was under no illusion that the movement in cinema was “real”; it was experienced only within the spectator as a kind of phantom. In the cinema itself, “there is no movement, no flux, no life in the mosaics of light and shadow which the screen always presents as fixed.” Christian Metz, on the other hand, in an early piece inspired by phenomenology on the impression of reality in the cinema, argued that there is no such thing as the reproduction, or representation, or illusion of motion; the appearance of movement *is* movement: “Because movement is never material but is *always* visual, to reproduce its appearance is to duplicate its real-

ity.”⁷ Movement is the annihilation of the distinction between object and copy and ensures the reality-effect of the cinema. Similarly, Deleuze has a stake in the idea that the cinema gives us real movement, for he allies himself with Bergson's claim that both movement and time constitute irreducible continuities. The investment in plenitude and consequent rejection of any underlying lack or absence are clear. For the moment, I simply want to reemphasize that a belief in the cinema's alliance with real movement rests on a denial of discontinuity, or of the significance of the distance separating photograms and incarnated in the frameline.

In their insistence upon the reality of cinematic movement, these theorists are duplicating the gesture of much of the earliest cinema—the constant reiteration of the cinematic fact of movement, of the capabilities of the machine. Much of this cinema could be characterized as the sheer celebration of movement for its own sake. This is particularly true of the earliest actualities, especially those of the Lumières, the most famous of which, *Workers Leaving the Lumière Factory* (*Sortie d'usine*, 1895), is content to chronicle the steady stream of workers emerging from the factory yard and moving toward and beyond the edges of the frame.⁸ The Lumières were adept at framing their subjects in depth so that the movement of the figures, traversing the frame diagonally, would be extended, would consume more time; examples include *Course en sacs* (*Sack Race*, 1896), *Stuttgart: 26^e Dragons. Sauts d'obstacles* (*German Hussars Jumping Fences*, 1896), *Arrivée en gondole* (*Gondola Party*, 1896), *Arrivée d'un train en gare de La Ciotat* (*Arrival of a Train at the Station of La Ciotat*, 1897), *Barque sortant du port* (*Boat Leaving the Harbor*, 1897).⁹ Other films extol movement through its insistent repetition, as in *Sauts au cheval en longueur* (*Leaping over a Horse*, 1897), *Baignade de négroillons* (*Negroes Bathing*, 1897). The anecdote detailing Méliès's fascination with the moving leaves in the background of *Repas de Bébé* (1895), rather than with the concerted (and centered) efforts of the parents to feed the child, is well known. Even in a film such as *Partie d'écarte* (*Friendly Party in the Garden of Lumière*, 1896) where the three cardplayers constitute an unusually static scene, excessive movement is displaced onto the waiter, whose near-hysterical reactions to the card game are left unexplained. That contortions of the body and especially its involuntary and violent movements were perceived as particularly cinematic is evidenced by one of the earliest films, Edison's Kinetoscope film *Fred Ott's Sneeze* (1894),

whose title encompasses its action. Similarly, the simplest of movements or gestures could be transformed into an event by being extended or prolonged (as well as performed by well-known theatrical actors). *The Irwin Rice Kiss* (Edison, 1896), which has become one of the most emblematic and recognizable of early films, receives the following description in the Edison catalogue: "They get ready to kiss, begin to kiss, and kiss in a way that brings down the house."¹⁰ The fascination with movement specific to the cinema and which differentiates it decisively from photography and painting is explicitly articulated in a group of films that represent the transformation of a still image (usually a painting) into a "live," moving human being: *The Artist's Dream* (Edison, 1900), *Animated Painting* (Edison, 1904), and *The Artist's Dilemma* (Edison, 1900).

The actualities that reproduce ordinary, recognizable movements and actions of the order of the everyday participate in a tendency that Jacques Aumont attributes to the Impressionist painters.¹¹ It is the attempt to seize the moment that flees and simultaneously to grasp it as a fugitive moment.¹² All the Lumières can do is multiply the number of such moments, seemingly indefinitely, and produce a series of catalogues containing 1,424 films, dividing the films ("vues") into such categories as "vues militaires," "vues comiques," "vues diverses." Theoretically, the topics are inexhaustible, but the Lumières' cinematic career was in fact quite brief. This approach to cinematic movement and time was historically short-lived, semiotically insufficient. The one-shot film, that slice of the apparently continuous indivisible unity of time and movement that displayed the capabilities of the cinema, was soon extinct.¹³

Up to now I have deliberately left vague the relation between movement and time. At the philosophical level, they are frequently addressed as inseparable issues, indeed as the same ontological problem. If Zeno can disprove the possibility of movement, he can demonstrate that there is no change, hence, effectively, no passage of time. When Bergson argues for the indivisibility of movement as a continuous whole, he is arguing the same for duration. Movement is often represented as the embodiment of time, and it is difficult to conceive of an access to time which is not mediated by movement or change (which itself seems ineluctably wedded to movement). The early cinema seems to corroborate these assumptions by foregrounding movement as the guarantee of its ability to capture or store time. Time be-

comes visible as the movement of bodies through space. Nevertheless, theoretically, time and movement are distinguishable as concepts. Deleuze, in what he claims is "the style of ancient philosophers," refers to time as the "number of movement";¹⁴ it is sometimes dependent upon what it measures (that is, movement) and sometimes an entirely independent instance. The classical cinema, according to Deleuze, maintains the subordination of time to movement, but the modernist cinema (primarily of the 1960s and 1970s) demonstrates that the cinema is capable of producing an image of pure time, liberated from movement.¹⁵ Whereas even classical cinema disengages movement from bodies through processes of montage and camera movement, the "primitive" cinema, in Deleuze's analysis, does not extract movement "for itself," but leaves it attached to "elements, characters and things which serve as its moving body or vehicle."¹⁶ Hence the primitive cinema could not realize the full potential of the cinematic representation of time.

Regardless of how one assesses Deleuze's history of cinema, he is not alone in noting that movement, time, and bodies are welded together in the continuity of the single-shot actuality. Given assumptions about the "real time" of the shot, this welding also implies a certain adherence of the film to reality. As André Gaudreault claims, "The filmic monstrator [the implied subject-producer of the display or exhibition in the actuality] . . . is 'captured' by the shackles of reality, of *his* reality: the apparatus of cinematography. That reality is one of twenty-four frames per second. The work of the camera-monstrator is necessarily of the order of the continuous," and "all monstrators have their 'noses' glued to the here and now of the 'representation.'"¹⁷

The single-shot actuality may not acknowledge the discontinuity upon which it is based, but it does embody a certain understanding of time and even a philosophy concerning its representation, that implicit in the development of the apparatus itself. For the true technological ancestor of the cinema, as Deleuze has pointed out, was not simply photography, but *instantaneous* photography, which allowed the registration of evenly spaced and sequenced moments. Cinematography made possible the synthesis through projection of such moments, but nevertheless depended upon their spacing and separate articulation. The apparatus here requires a certain leveling; time is a series of equivalent and equidistant instants (twenty-four frames per second) subjected to no hierarchy whatsoever. Cinema deals with

the “any-instant-whatever.”¹⁸ Although Bergson asserts the “impossibility of real instants” in his own philosophy, he acknowledges that there have been historically two different ways of erring in the attempt to define or specify the instant. The ancients believed in “privileged” instants, characteristic and essential poses. These were the embodiments or actualizations of a transcendent form, eternal and immobile. The transitional moments between privileged ones were of no interest. Time was punctuated by a hierarchy in which certain moments were privileged as the actualization of an ideal or essential form. The moderns err, on the other hand, by refusing to hierarchize instants (as well as by proclaiming their very existence), by assigning them all to the same banal ontology. Instantaneous photography, for Bergson the quintessential apparatus of modernity, isolates “any moment” and “puts them all in the same rank.”¹⁹ In the thought of the ancients, significance was transcendental and preceded embodiment; in that of the moderns, significance emerges from an immanence with no hierarchical guarantees. An instant may *become* remarkable or privileged, but at its origin it is as ordinary and banal as any other. Deleuze allies this thinking with that of modern science, which understands time as a “mechanical succession of instants,” rather than in the ancient fashion as “the dialectical order of poses.” Bergson rejects both attitudes as a misrecognition of the true (that is, nonspatial) character of time, but Deleuze claims that the philosophy of any-instant-whatever, built into the technology of the cinema, is in line with Bergson’s theory of time insofar as it allows for thinking the production of the new, which can be connected to *any* moment. Meaning is predetermined not in ideal forms, but in a process of emergence and surprise. According to Deleuze this is a “complete conversion of philosophy” and goes by the name of modernity.²⁰

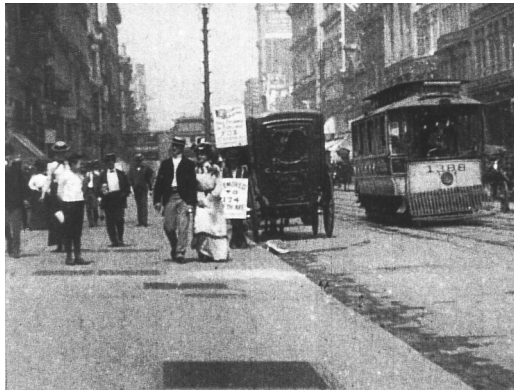
Early actualities seem to confirm this thinking about time: “any-instant-whatever” becomes cinema’s proper topic. Although the placement of the camera may be precisely calculated and the recorded activities foreseen or tightly regulated, these films depend on the fascination associated with the camera’s ability to “catch” moments, to itself be surprised by meaning. Dai Vaughan’s homage to the Lumières rests on an affirmation of precisely this kind of unprecedented alliance between representation and unpredictability. Deleuze aligns the cinematic apprehension of temporality with movements in the other arts (ballet, mime) toward the end of the nineteenth century

that exploit accidents of the environment. Significance is not predetermined in an ideal form, but emerges out of the accident; it is variable and unpredictable. An early actuality, *What Happened on Twenty-third Street* (Edison, 1901), explicitly plays out this scenario, tracing the emergence of meaning.²¹ The single-shot film begins as an almost unreadable frame encompassing the varied activities on a city street and the sidewalk next to it (Figure 6.1). People and carriages pass by in no particular order, some of them glancing at or acknowledging the presence of the camera, and the film at this point resembles the “purest” of actualities, in which the camera is simply aimed at a street scene and allowed to record whatever happens. But gradually a man and woman, walking together, emerge from the mass of details and walk toward the camera (Figures 6.2 and 6.3). As they cross a grating, a gust of air blows up through it and lifts the woman’s skirt (Figure 6.4). She is momentarily embarrassed, but recovers and laughs (Figure 6.5). It is perhaps not accidental that the meaning which emerges from the chaos of detail in the street scene is an eroticized one, calling for a voyeuristic gaze; I will return to this question later. What I would like to emphasize now is that the film effectively demonstrates that contingency presupposes a certain originary evacuation of meaning. Moreover, the early cinema gives the spectator the opportunity of witnessing the ceaseless production of meaning *out of* contingency.

Hence the early cinema is very much about instants and their accountability with respect to meaning. Aumont also invokes a binary opposition in conceptualizing the instant and links it to a history of painting, photography, and cinema. Gotthold Ephraim Lessing’s notion of the “pregnant instant” in painting—the best instant, the most significant, typical instant—is opposed to the any-instant-whatever that Aumont associates with Impressionism and photography. The pregnant instant is linked to processes of symbolization and the production of meaning, while a focus on any-instant-whatever is evidence of the lure of realism and of “revelation.” But the term *pregnant instant* is, according to Aumont, an oxymoron. One cannot effectively conjoin instantaneity (or the authenticity of the event, the real) with an immediacy and fullness of meaning, for “meaning has no place in the real.”²² Yet Aumont claims that the term is particularly useful, for it perfectly describes a transitional moment in the history of painting—the movement from a form of representation that maximizes “pregnancy” to one that maximizes the instantaneous and the accidental. The credo of the “pregnant mo-



6.1 *What Happened on
Twenty-third Street* (1901).
Edison.



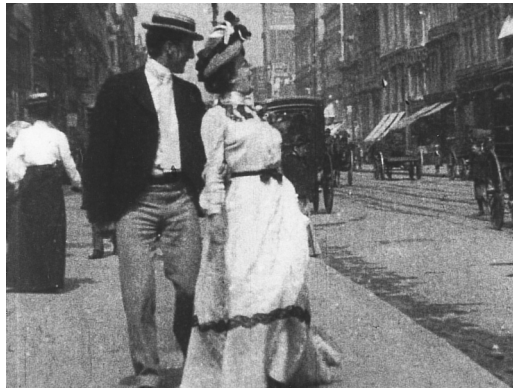
6.2 *What Happened on
Twenty-third Street*.



6.3 *What Happened on
Twenty-third Street*.



6.4 *What Happened on
Twenty-third Street.*



6.5 *What Happened on
Twenty-third Street.*

ment” is antithetical to tendencies within Impressionism, which cultivates the values of the ephemeral, the circumstantial, and sensation as opposed to meaning. Early photography may strive to transform any-instant-whatever into a pregnant instant (a process that has more to do with “revelation” than with symbolization), but the tendencies that dominate modernity circulate around impossible desires—how to represent the unrepresentable (as in Turner) or the impalpable (light for the Impressionists), and how to fix the ephemeral. The persistent conflict between meaning and contingency which haunts the efforts of photography and painting in relation to the representation of time seems to be neutralized by the camera, which replaces choice and intentionality with an automatic inscription of duration.

Whereas painting has to struggle to produce the sign of time, temporality

is one of the signifying materials of the cinema, a part of its experience for the spectator. However, there is a sense in which the very concept of the instant is inappropriate for defining the cinema, which always deals with an extensive duration. Even the shortest shot traces a process in time, and a freeze frame dictates the duration of its own reception. The instant, properly speaking, belongs to photography and to the individual film frame, which is never seen as such by the spectator. A single shot inevitably produces the effect of temporal continuity and, hence, of “real time.” In theories of the early cinema, this adherence to continuity is almost always perceived as a limitation that must be overcome before the cinema as a significant form of representation can truly emerge. Gaudreault refers to the continuity of the apparatus as a form of “shackles.” Deleuze believes that the essence of the cinematic is not visible until time can be disengaged from the movement of bodies within the diegesis and *articulated* through montage. Ultimately it is editing, the possibility of a *cut* in the temporal and spatial continuity of the shot, that is fetishized as the semiotic imperative of the cinema. For general cultural theorists such as Walter Benjamin and Siegfried Kracauer, the cut was *the* incarnation of temporality in film, and it constituted the formal response to the restructuring of time in modernity.

Hence a hole or breach is perceived (at the levels both of filmmaking and of its theorization) as crucial to cinematic signification. This hole is not located, as it might be, in what is lost at the edges of the frame, or in the lack of a third dimension or color, but as the possibility of an interruption in the linear forward movement of the film strip, of its *rearticulation* in editing. According to Gaudreault, the filmic monstrator, subjected to the tyranny of temporal continuity in the single-shot actuality, does not have the option of “opening a temporal breach,” an action that would permit the inscription of a reflection on the narrated world, a “filtering” through the eyes of a narrator.²³ For Gaudreault, it is editing which allows the inscription within film of “temporal *différance*”; the present tense of the spectator’s reception is conjoined with the past tense of a filmic narration. The spectator sees *now* what the narrator *has seen* before and rearranged, rearticulated for the benefit of the spectator.

Despite the antiteleological thrust of current film historiography, its desire to grant to early cinema its own autonomy in relation to later “developments” in classical narrative cinema, the temporal continuity of the single-

shot actuality continues to be conceptualized as a limitation, as a primitive moment. Hence the literature stresses the necessity of the cinematic discovery of (at least) two temporalities, two types of movement. For Deleuze, camera movement and montage provide an alternative, specifically cinematic temporality, which liberates film from the shackles of a temporality enslaved to the movement of bodies and things within the frame. Gaudreault also outlines the historical necessity of a “double mobility.” He claims that, in the single-shot film, “The double mobility which characterizes film narrative—mobility of objects depicted and motility of time-space segmentation—*had yet to materialize completely*. All that was known was the mobility of objects depicted in the frame, a mobility made possible by the new invention which could seize, fix and reproduce movements of beings and things.”²⁴ This seems to suggest that the alignment of film with a pure storage/recording of time was at the same time desired and spurned. Theories of double mobility or double temporality in the cinema respond to the necessity of accounting for both film’s indexicality and its possibility of articulation (of syntax). On the one hand, the cinematic image appears as the imprint of time, its automatic rendering and recording. But the implicit history in the official histories dictates the intolerability of this state of affairs; plenitude poses a threat. A gap or interval is required and is found in the form of editing. On the other hand, the solution to the threat of the overpresence of the image—editing—generates its own anxieties about discontinuity and absence. As a reinscription of the gap between film frames, editing potentially constitutes a persistent reminder of the abyss of darkness that subtends cinema.²⁵ Aumont claims that the fear of overpresence outweighs the fear of a hole or gap because it is “easier, mentally *and* visually, to fill a hole than to assimilate [*résorber*] a too-full [*trop-plein*].”²⁶ But I think it would be more accurate to say that it is in the tension between the fear of surfeit and the fear of absence that the specificity of the cinema’s inscription of time lies.

Nevertheless, it remains clear that editing—as the possibility of departure from temporal and spatial continuity—is consistently perceived as the *sine qua non* of cinematic signification. Aumont maintains that the “interval” (usually conceived as the “visual distance maintained between two shots”) is what permits the cinema to make of time a formal material.²⁷ Kracauer finds the spatial continuity of photography abhorrent and fully in collusion with

the temporal continuity embraced by historicism. What both disallow are the gaps conducive to the production of significance and authentic memory. However, the blanketing of the world by photographs evacuated of meaning is countered, in a utopian discursive moment toward the end of Kracauer's essay, by an appeal to film's "capacity to stir up the elements of nature" by combining "parts and segments to create strange constructs." The fact that editing in itself does not provide an authentic or true arrangement, but instead, in the manner of dreams, reveals that "the valid organization of things remains unknown," indicates that for Kracauer it is the process of editing itself which can counter the semiotic-historical deficiencies of photographic continuity.²⁸

Given the critical emphasis on the interval, the gap and the production of temporalities freed from the movement of objects and bodies in the frame, it is easy to see why film historians dwell on the vicissitudes of editing in the earliest years of the cinema. For the struggles to produce legibility and logic in sequencing shots reveal the complexities of creating an articulated time that does not undermine the authenticity/authority of recorded time—the "real time" of the shot. The joining together of two shots takes place very early in film history, but the implications of that act have varying levels of significance. In early actualities, such as *Electrocuting an Elephant* (Edison, 1903) and *The Burning of Durland's Riding Academy* (Edison, 1902), cuts (or camera stoppages) represent a simple ellipsis, eliding "uneventful" time, or they enable the presentation of multiple views of the same "event" (as in *Burning*). The cut (or camera stoppage) in some execution films (*The Execution of Mary Queen of Scots* [Edison, 1895], *Execution by Hanging* [Mutoscope/Biograph, 1905]) is designed precisely *not to be seen*, so that the substitution of a dummy for the real person in the execution will be invisible to a duped spectator. Here, the contents of the image are changed, but not its framing. This is a tactic used extensively in the trick film and especially in Méliès's work, where what is at stake is the fascination associated with sudden appearances and disappearances of bodies and things. Also, it is frequently noted that editing in the early cinema assumes the role of a simple addition or accumulation of shots constituted as tableaux, shots that could easily stand on their own as independent films. The instability of such editing is demonstrated by the fact that exhibitors were given the option of renting single shots.

Yet there is also a pronounced tendency to utilize editing in such a way that cuts will be technically visible but not disruptive, and their location stable and necessary in the effort to build a recognizable filmic world. What I am interested in isolating and examining here are three logics of early film editing that involve the deliberate construction of a diegesis, the concerted effort to formulate a coherent relation to filmic space and time. These logics are not exhaustive of such efforts, but historians nevertheless return to them again and again as exemplary moments in the development and stabilization of a cinematic syntax. I am interested in analyzing these moments specifically in relation to their assumptions about temporality, about the potential form of a cinematic time. The three editing practices involve: the logic of repetition (in which a change in the position of the camera instigates a repetition of a narrative event—*The Life of an American Fireman* (Edison, 1903) is the most frequently cited example); the logic of the chase (the insistent linearization of time); and the logic of parallel editing and suspense (the dramatization of time). Ultimately I will examine the relation of these logics to the difficulties posed by Zeno's paradoxes in their reemergence at the turn of the century.

In retrospect, the logic of repetition strikes us as closer to an antilogic, or the violation of a coherent set of rules for the construction of filmic space and time. These rules—those of classical continuity editing—did not emerge until much later than the films considered here, and therefore it is not the question of “violation” which is at stake.²⁹ The most famous example is Edwin S. Porter's *The Life of an American Fireman*. The overall organization of the film is that of linear narrative, but the crucial scene—in which the mother and daughter are rescued by the firemen—is repeated in toto, represented from two different points of view. In the first scene, the camera is located “inside” the house and records the mother waking, running to the window, and collapsing on the bed. A fireman enters the bedroom, breaks the window, and carries the mother out the window and down a ladder. A few seconds later he returns up the ladder to get the child. This scene is repeated from the “outside” of the house, the fireman descending the ladder with the mother, the mother pleading with the fireman to save her child, and the fireman returning to get the child. In each case, any elapsed time is situated in the “other,” unseen space. Although this is the most elaborate example of temporal repetition in early cinema, there are other examples clus-

tered around the same year, 1903. *Next!* (Mutoscope/Biograph, 1903) is a short, two-shot skit that takes place in a barbershop. Two clownish-looking patrons (Alphonse and Gaston) graciously defer to each other, bowing repetitively as the barber calls for the next customer. Impatient and frustrated, the barbers and the other customers finally throw the two clowns out the window, one after the other. The next and final shot repeats the same action (the two being thrown through the window) from another camera position, outside the barbershop. In another Porter film, *The Strenuous Life, or Anti-Race Suicide* (Edison, 1904), a cut-in to a closer shot of the father holding and weighing his new baby initiates a replay of a previous action. In each of these examples, temporal repetition seems to compensate for spatial dislocation. The spectator's "anchor" would be the familiarity/recognizability of a filmic time already "lived."

Around 1903–04, concerns about the stability of point of view seem to outweigh the disadvantages of temporal repetition in films that are in other respects quite linear. Noël Burch astutely links this temporal repetition to the specificity of the spectator's relation to the screen in the early cinema. Referring to *The Life of an American Fireman*, he claims that

the fact that once these two shots were filmed, it was decided to connect them in a manner implying an obvious non-linearity rather than disturb the unity of the spatial viewpoint, seems to me to say a good deal about the *alterity* of the relationship these early films entertained with the spectators who watched them. Does it not suggest that the feeling of being seated in a theatre in front of a screen had, for spectators then, a sort of priority over the feeling of being carried away by an imaginary time-flow, modelled on the semblance of linearity which *ordinary time* has for us?³⁰

The logic of temporal repetition implies a resolute literalism in the cinematic depiction of time. In a medium so strongly applauded for its ability to store or record time, it becomes difficult to think time as figurable, manipulable. The linearity and irreversibility of film would seem to make it most resistant to the depiction of simultaneity when different spaces are involved. In the first shot of *The Life of an American Fireman*, a fireman dozes at his desk. A balloon insert appears in the upper right corner of the frame, and

within it the mother puts her child to bed. This has been interpreted in various ways as the fireman's dream about his own family or as the representation of simultaneous events in different spaces. Since the mother and child are never connected in any other way to this particular fireman, the argument that the superimposition depicts simultaneity strikes me as more plausible. Other films of the period, such as *The Story the Biograph Told* (Biograph, 1903), also resort to superimposition for the depiction of simultaneity. In this instance, the resolute literalism of the representation of time is apparent. What happens simultaneously in the narrative happens simultaneously in the image. Filmic time and diegetic time coincide. There is a certain spatial violence enacted here, and the coherence and contiguity of space are sacrificed to a literalism of time. Parallel editing is the historical "solution" to the problem of representing simultaneity, but in order for it to be viable, succession must be accepted as the signifier of its opposite, simultaneity. The cinematic troping of time begins.

For the most part, visible time in the cinema is equal to "real time," and any manipulation or troping of time takes place in the invisible realms of off-screen space or the interstices between shots. (Fast motion, slow motion, the freeze frame, and other distortions of time become, precisely, *special effects*, relegated to the marginal status of heavily coded—and rare—moments.) In *The Life of an American Fireman*, time is in fact condensed whenever events take place off-screen. In the "inside" view, the time that elapses between the moment when the fireman disappears down the ladder with the mother and the moment when he reappears to save the child is hardly sufficient to take the mother to safety and receive her pleas for the rescue of her child (events that are depicted in the "outside" portion). As Charles Musser has pointed out, "In keeping with theatrical conventions, whenever actions take place off screen, time is elided."³¹ Time is manipulable when it is invisible. But *The Life of an American Fireman*, by repeating the scene, reveals what was "off" and formerly condensed, and "reexpands" it. From the point of view of later developments in continuity editing, this could only be perceived as a "mistake" or a "violation."

Although temporal repetition in *The Life of an American Fireman* and other films of the period certainly reveals spatial anxieties and apprehensions about the mobility of the camera, and hence the imaginary mobility of the spectator, it also very crucially indicates the forcefulness of the represen-

tational struggles over cinematic temporality. The manifest project of modernity—and of the cinema in the wake of Marey and Muybridge—is to make time visible, representable, to store and, hence, to defeat time as relentless passage. But visible time in the cinema is yoked to the body, movement, space, and the potential excesses of “real time.” The primary viable articulation of time takes place off-screen, between frames, in darkness. It is in the intervals, invisible aspects, blind spots, the gaps, that time becomes accessible to manipulation, narrativization, and ultimately commodification. The cinematic production of temporality, for the most part, goes unseen, is naturalized through a logic of “real time.” The representational struggles over temporality are fought on the terrain of the visible versus the invisible.

In *The Life of an American Fireman* and *Next!* what is “lost” to invisibility, and then “filled in” by the temporal repetition, is the depth of the image, its missing dimension. The window on the back wall of each set suggests the existence of an unseen narrative realm, but even more insistently, the fact that characters *cross* that semiotic barrier, transgress the spatial/temporal limits of the frame in an action central to the narrative, makes the invisibility of the “beyond” of the frame intolerable. It is made visible at the cost of a temporal jolt, a redundancy. Such a logic cannot survive, because it violates the perceived alliance of the medium with temporal irreversibility (see Chapter 4). The chase film, on the other hand, yokes invisibility to a logic of succession, to a horizontal imperative of linearity. Here, however, the desire to make and keep everything visible is also strongly evident, this time in the convention whereby all characters are allowed to leave the frame before the cut to the next location. An empty frame fills with pursued and pursuers, is evacuated, and the cycle continues. The cut is made when there is no longer anything to be seen. As numerous critics have pointed out, editing in the early cinema is often a matter of the simple juxtaposition of autonomous tableaux, as in *The Passion Play of Oberammergau* (Eden-Musee, 1898). Given what Noël Burch refers to as the “autarchy” and unicity of the frame, the space and time of the first shot do not interpenetrate the space and time of the second; any connections between the two are external. The logic is that of the sheer addition or accumulation of views. The chase film introduces a tension and a direction to filmic temporality and vectorizes space. Time is structured through anticipation and delay. *Personal* (Biograph, 1904), in which a dandy advertises for a wife and, overwhelmed with applicants for

the position, is pursued down a city street, through fields, down a country road, across a fence, and so on, until he is finally “caught” by one of the women (who pulls a gun on him), is in many ways paradigmatic of the genre (extremely popular, it was remade at least three times).³² Each shot is set in a different location but encompasses the same action—pursuers chasing the pursued. The spaces are not connected geographically or filmically in any way other than through their similar functions as the field or background traversed by the characters. The same general logic informs films such as *The Lost Child* (Biograph, 1904), in which a large crowd (including an old man in a wheelchair and a woman with a stroller) chases a man mistakenly thought to have abducted a child, and *Getting Evidence* (Edison, 1906), in which a photographer attempts unsuccessfully to photograph an “illicit” couple in a variety of comic situations and, once he obtains the desired photograph, is subject to an extended chase.

In all these films the movement of characters is either toward or away from the camera, extending the amount of time they are visible within a single frame. And although the chase film is often described as the insistent linearization of filmic temporality, there is something curiously static about these films. Aside from the beginning and ending shots, the order of shots is entirely arbitrary. It does not matter whether the dandy in *Personal* is chased across a fence before or after he is chased down a road. The middle of the film is basically constituted by repetition masquerading as difference. The chase stretches, elongates, fills the time of filmic representation, often to the point of monotony. Burch claims: “And this was the role of the chase: to extend the film experience, to initiate a certain ‘imaginary’ production of duration and succession exploiting an off-screen space which although it was still amorphous would eventually make possible the diegetic production characteristic of the institution.”³³ Since repetition, duration, and succession were the laws of this logic, it was not in fact fully confined to the chase film, and characterized films such as *Rescued by Rover* (Hepworth, 1905), *Rube and Mandy at Coney Island* (Edison, 1907), and *Laughing Gas* (Edison, 1907), although these films lack the tension of heavy pursuit. In *Laughing Gas* (Edison, 1907), a black woman named Mandy is given laughing gas by a dentist when her tooth is pulled. She goes on, in a succession of different locations and situations (the subway, the street, a court, a church), to infect everyone she meets with her unstoppable laughter.

There is something excessive, almost hysterical, about the films that deploy this logic. Nearly all of them are comic, but the comedy seems only to intensify the anxiety they reveal. In *Laughing Gas*, what is clearly at stake is a fear of contagion linked to racial difference. In *Rescued by Rover* and *The Lost Child*, it is the safety and security of the bourgeois nuclear family that are threatened from without (by a gypsy or stranger), and the figure of the lost child acts as the emblem of loss in general. In *Personal*, it is femininity itself which constitutes a threat through the incessant multiplication of the figure of the demanding and quasi-hysterical potential wife. A mob of women repeatedly fill and empty the frame, and the dandy becomes a besieged and sympathetic character. There is a certain hysteria of sexual difference here, the inscription of an excessive heterosexuality gone awry. Lubin's remake of *Personal*, titled *Meet Me at the Fountain* (1904), comments upon the insistency of this heterosexual matrix by including, as one of the pursuing "women," a well-known female impersonator (Gilbert Saroni), who, though always the last to enter and exit the frame, "gets" the man in the end, a scene that is constituted by a closer shot in which the female impersonator caresses and kisses the dandy, who wipes his mouth with his handkerchief. The "joke" assumes heterosexuality as a norm and as normal. Similarly, *Getting Evidence* circulates around heterosexuality, the sought-after evidence constituted by photographic proof of the intimate relation of a man and a woman. The constitution of a cinematic syntax and the incessant inscription of heterosexuality as norm are in collusion.

The structure of the chase film partakes of the logic of the series, subsuming individual irregularity beneath the rule of aggregate regularity, and allying itself with a growing social emphasis upon statistics and probability. Each shot generates a sense of difference but reiterates the same dilemma: the pursuers chase the pursued, traversing the depth of field, until some arbitrary moment of closure, usually catastrophic, ends the series. And the series is repeated again in the next film, with the same expectations, the same use of delay to construct filmic duration. As Ian Hacking has demonstrated, during this period, correlation began to displace causality, and "A story of the erosion of determinism is also an account of the invention of normalcy."³⁴ The isolation of causal factors gives way before the observation of statistical regularities and processes of normalization. The films use the combination of individual irregularity with aggregate regularity in order to

buttress the norms of family, heterosexuality, and whiteness, all of which are perceived as threatened at the turn of the century. Filmic duration is organized to suggest that each shot entertains differences, implies a progression, when the procedure involved is instead one of static addition, accumulation. As Hacking points out, the ideology of normalization deploys simultaneously two different understandings of the “normal”—“what is,” the usual, the average, the typical (which may be subject to deviation, deflection), and, second, what ought to be, what we should strive for, the ethical standard of the “norm.” Such a double temporality implies a dual understanding of progress as both the restoration of an ideal average (for example, “lost health” as restoration of the “norm”) and the movement toward a future goal (normalization). Both are curiously static teleologies. The insistent linearization usually attributed to the chase film, which conceals an equally insistent redundancy and repetitiveness in the service of normalization, invests in this ideology of progress.

While the first two logics of film editing have been perceived as stages, as evidence of a certain stylistic “primitiveness,” the use of parallel editing is frequently treated as the telos of early cinema, the attainment of a fully and specifically cinematic discourse. Parallel editing successfully eroticizes time, injects it with desire, expectation, anticipation, and displaces the spectatorial time of viewing by contributing to the construction of a “lived,” imaginary temporality. Parallel editing is generally defined as cutting alternately between two scenes that are assumed to have some form of relation with each other and are usually understood as taking place simultaneously or nearly simultaneously. Simultaneity is signified by succession rather than by the spatial disruption of the image through balloon inserts or superimposition. Filmic time no longer matches, in a one-to-one correspondence (as in the single-shot actuality so long dominant), diegetic time. Spectatorial time—the time of sitting in the theater—is emptied and replaced by the cinematic articulation of a temporality whose strategies are concealed by an appeal to the grounding referentiality/indexicality of its base.

In parallel editing, the relations between the two shots are no longer the benign ones of addition or accumulation, but are intensified by the interpenetration of anxiety, desire, or horror. This is most evident in the parallel editing that links shots of an imperiled heroine or heroines, or family, with shots of the potential rescuers (*The Lonely Villa* [1909], *An Unseen Enemy*

[1912], or *The Lonedale Operator* [1911]), but is also characteristic of its discursive or metaphorical activation, as in the comparison of rich to poor in *A Corner in Wheat* (1909) and *The Usurer* (1910) (all Biograph). D. W. Griffith's significance here lies in his extensive use of parallel editing and his insistence upon making the articulation of time as crisis crucial to cinematic signification. *The Fatal Hour* (1908), the first film in which Griffith used extensive parallel editing,³⁵ ends with a scene in which shots of a female detective immobilized in front of a gun timed to shoot her when the clock strikes twelve are intercut with shots of the police coming to the rescue. They, of course, reach her in "the nick of time."

The yoking together of noncontiguous spaces through parallel editing forced a certain denaturalization of the filmic discourse. It required the spectator to accept enormous leaps in space and to allow the disfiguration of continuous time, its expansion or contraction. As Gunning has argued, the temporal simultaneity signified by parallel editing demands an abstraction that was countered by the incorporation of recent technology—the telephone, the telegraph, the railroad—into plots, in order to "naturalize film's power to move through space and time."³⁶ Editing borrowed the authority of the telephone to rationalize the instantaneous movement from site to site effected by the cut. Telephones are indeed ubiquitous in the films of this period. In *An Unseen Enemy*, two orphaned sisters are trapped in a room by their "slatternly" maid and her accomplice, who attempt to steal the contents of the father's safe. It is only when one of the sisters manages to reach the telephone that she is able to summon her brother to help them. In *The Lonely Villa*, Gunning's primary example, a bourgeois home is threatened by three itinerants, and the wife, hiding in an interior room with her three daughters, is able to alert her husband to the danger when he phones her. In both of these examples, a telephone call initiates a sequence of parallel editing chronicling a rescue. In *The Medicine Bottle* (1909), the telephone itself constitutes the solution to the problem (a woman has mistakenly instructed her daughter to administer poison instead of medicine to her sick grandmother), and the only possible hindrance or delay is in the telephone system itself (the switchboard operators gossip instead of attending to their duties). In *Death's Marathon* (1913), when a friend is too late to prevent her husband from killing himself, the wife "witnesses" the suicide over the telephone after a prolonged sequence of parallel editing. *The Lonedale Operator* combines railroad and telegraph in its narrative of separation, threat, and rescue.

As Gunning astutely points out, the motivation for this type of parallel editing is most frequently the desire to overcome a distance or physical separation, to “join . . . loved ones.”³⁷ Suspense is predicated on absence or separation and driven by an external threat to the home, the family, the woman. The gap between shots mimics the gap constitutive of desire. Suspense in the cinema, as Pascal Bonitzer has extensively demonstrated, is on the side of invisibility, and depends upon the activation of off-screen space, or the “blind spot.”³⁸ In parallel editing, when shot B is on the screen its legibility is saturated by the absent presence of shot A, and vice versa. Duration is energized by invisibility, by the inability to see all. It is not accidental that the “invention” of parallel editing and an increased exploitation of the frame as the threshold of an unseen yet semiotically dense space should coincide. In many of Griffith’s last-minute-rescue films, the women are encased in a room in a house or in a series of adjoining rooms, an architecture that allows for an intense awareness of encroaching danger “just beyond” a frame which coincides with and rearticulates the walls of the room. In *The Lonely Villa*, when the three vagabonds break through the door of the first room in which the women are hiding, they move to a second, adjacent room, and the process is repeated. In *An Unseen Enemy*, the contiguity of the room in which the orphaned sisters hide and that in which the maid and her accomplice attempt to break into the safe is vividly incarnated in a small hole connecting them through which a hand with a gun appears to threaten the sisters. Jacques Aumont’s reading of this spatial arrangement is provocative:

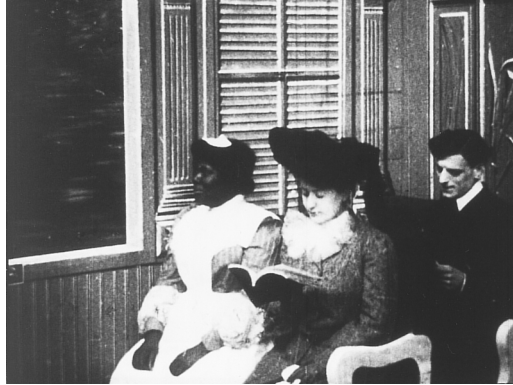
It has been remarked, in the context of *Intolerance*: in Griffith’s cinema, “partir c’est toujours mourir un peu” (“to leave is always to die a little”) and leaving the scene signifies at least potentially the death of the character. What closes the scene, via the frame, is in a sense this threat of death, always implicitly and metaphorically proffered from beyond the frame. The revolver of *An Unseen Enemy* can, God knows, be read in more ways than one, but . . . can we not still see in it the literal agent of this threat, openly articulated in this way?³⁹

Time, death, and invisibility are welded together at the edge of the frame and between shots, in the unseen space that makes it possible for the cinema to say anything at all. Perhaps this is why suspense (and its deployment in parallel editing) was so crucial to the forging of a cinematic syntax in mo-

dermity. Fatality is the perpetually deferred that drives the linear movement, and, as Bonitzer has shown, it is the “image of the worst” that governs the progression of shots in suspense, as a system.⁴⁰

This fatality most frequently takes the clichéd form of the woman in distress, the imperiled heroine. *The Fatal Hour* is only the most explicit instance of the early cinema’s tendency to yoke together the figure of the threatened woman and a discursive insistence upon temporality. Time itself seems to menace the woman, and it is in “the nick of time” that she is saved. In *The Fatal Hour*, it is the Law that comes to the rescue, but more frequently it is a lover or a husband (as in *The Lonedale Operator* or *The Lonely Villa*), so that, in effect, it is the mechanism of heterosexuality which ensures her salvation. Closure is effected by the embrace or the kiss, over and over again, as the resolution to the problem of an ending in a cinema that affirms time as continuity. In *The Lonedale Operator*, as Raymond Bellour has demonstrated, the logic of the parallel editing is basically the logic of an alternation between “he” and “she.” The film begins with a scene in which the operator playfully parries the attentions of the engineer. It is only after their separation and the ensuing threat to her well-being that they are successfully reunited and embrace at the end of the film. The “thrilling of intelligibility” that Barthes associates with suspense is in the service of the confirmation and reconfirmation of heterosexuality, which in turn lends its “intelligibility” to the cinematic desire to overcome distance, separation, the inevitability of absence.⁴¹

The compulsive replaying of the scenario of heterosexual union exploits its mishaps and potential misses, energizing time and injecting it with a direction. Anxieties about the rationalization of time in modernity, about the confrontation with racial otherness, about emerging instabilities of gender identity, can be allayed by the insistent repetition of an imperative, normative, and fully realizable heterosexuality that overcomes division itself (a difference and a division that the cinematic medium itself has a stake in overcoming in the production of movement). Perhaps this is why the kiss has constituted such a crucial semiotic event in the cinema, from Edison’s 1896 *Irwin Rice Kiss* to *The Lonedale Operator*, and throughout the period of the classical Hollywood narrative. In this respect, the 1903 *What Happened in the Tunnel* (Edison) is quite telling.⁴² The film begins with a shot of a middle-class woman and her black maid sitting together in a train compartment



6.6 *What Happened in the Tunnel* (1903). Edison.



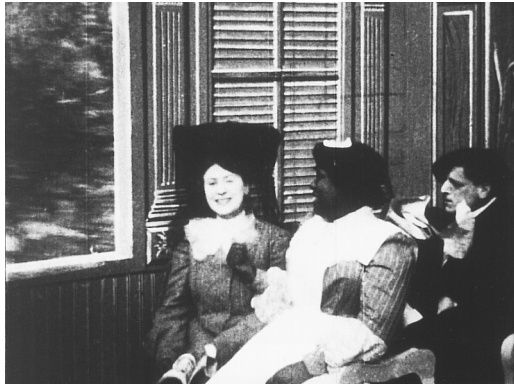
6.7 *What Happened in the Tunnel*.

(as a matte shot of a changing landscape on the window produces the illusion of the train's movement) (Figure 6.6).⁴³ A man sitting behind them is energetically flirting with the white woman (Figure 6.7). The train enters a tunnel (signified by a cut to black), and when it emerges (in a return to the first shot), the man finds that the women have changed places and he is kissing the black maid (Figure 6.8). Both women laugh heartily as the embarrassed would-be lover hides behind his newspaper (Figure 6.9).

Fashioning itself in the "what happened" mode of *What Happened on Twenty-third Street* (Edison, 1901), the film's pretense is that of having "captured" a particularly comic moment hinging on sexual difference. But in *What Happened on Twenty-third Street* contingency is conquered in a single shot—the revealing of the female body as spectacle binds the apparent anar-



6.8 *What Happened in the Tunnel.*

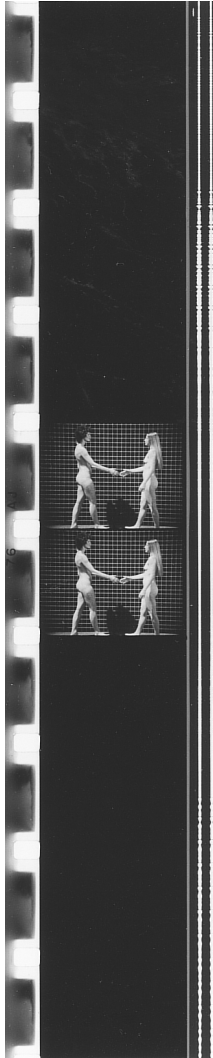


6.9 *What Happened in the Tunnel.*

chy of details in a focused look. *What Happened in the Tunnel*, on the other hand, utilizes editing, at a very basic level, in order to activate blindness, invisibility, as a structuring component of the cinema's articulation. On the one hand, the cut to blackness that bisects the film acts as the reinscription of the gap between frames and of the darkness built into cinematic duration through the operation of the shutter. On the other hand, that gap is recuperated for meaning through its narrativization, its diegetic incarnation as the passage through a tunnel. Nevertheless, the gap continues to embody a deeply felt threat, that of miscegenation, an "improper" sexual relation, which the film fends off through its transformation into comedy (although the laughter provoked is, no doubt, an anxious one). The kiss has been misplaced; it has gone awry.

The silent cinema did not fully master the darkness subtending the medium until the cranking of camera and projector was fully regularized through electrical power and the three-blade shutter introduced.⁴⁴ It is only then that the phenomenon called “flicker” could disappear. In the 1960s, the avant-garde film movement known as structural or materialist film refocused attention on the phenomenon of flicker in its investigation of the ontological implications of the material base of the medium—hence the short-lived genre of the “flicker film.” Thom Andersen’s remarkable documentary, *Eadweard Muybridge, Zoopraxographer* (1974), is very much in the spirit of this celebration of filmic materiality. Here Muybridge emerges as the heroic figure in the passage from photography to cinema, from static immobility to the successful reproduction of time and movement. Obsessed with movement, the camera is rarely still in Andersen’s film, even when it confronts the enormous range of Muybridge’s work in still photography, where the camera insistently moves into or out from the photographs, isolating details or placing them in context. Andersen also incessantly transforms Muybridge’s sequential photography into cinema, presenting it in a way that, given advances in cinematic technology, as Andersen points out, Muybridge himself could never have seen it. In *Eadweard Muybridge, Zoopraxographer*, athletes jump over fences repeatedly, women sweep floors and walk up stairs over and over again in a vertiginous display of, and incantation to, movement.

In a very striking sequence toward the end of the film, Andersen returns to the thematics of Zeno’s paradox and proposes the cinema anticipated by Muybridge as the solution. The sequence begins with a statement in a male voice-over, “But in Muybridge’s work, photography in its passage to cinema overcame a philosophical obstacle,” which precedes a cut to black. The shot that accompanies the discussion of this philosophical obstacle is a restaging of one of Muybridge’s scenarios (shown earlier in Andersen’s film) in which two women walk toward each other and kiss. The scene is clearly a “rewriting” of Muybridge—the two women are naked, they “perform” against a background constituted as a grid of white lines over black, and their gestures mimic precisely those of the women in Muybridge’s sequence. The flicker effect of this scene is at first pronounced, so that the spectator is intensely aware of the intervals of blackness between illuminations of the image (Figure 6.10). But as the scene progresses, the camera moves forward toward the women, and the flicker gradually decreases (Figures 6.11 and 6.12), until, at



6.10 *Eadweard Muybridge, Zoopraxographer* (1974).
Thom Andersen.



6.11 *Eadweard Muybridge, Zoopraxographer.*



6.12 *Eadweard Muybridge, Zoopraxographer.*



6.13 *Eadweard Muybridge, Zoopraxographer.*

the moment of the kiss, it disappears and the sequence becomes cinema (Figure 6.13). The overcoming of the darkness and the lost time it signals are coincident with the kiss. It is worth quoting the voice-over at length:

The modern motion picture camera bisects time into equal moments of light and darkness,⁴⁵ their duration regulated by the constant rotation of the shutter. Between each frame, when the shutter closes over the lens as the strip of film is repositioned, there is a moment of darkness, a fragment of time which is not recorded. Time, like any continuous quantity, is infinitely divisible. It cannot be reconstituted in its unity by a mechanical instrument which bisects it finitely. [At this point, the women touch hands.] So we might question the possibility of cinema, asking how can we recapture motion in a finite number of still pictures just as Zeno asked, how can we cross space in a finite number of movements, since it is infinitely divisible? The solution to this paradox lies in the persistence of vision, metaphorically described in the fifteenth century by Leonardo da Vinci, "Every body that moves rapidly seems to color its path with the impression of its hue. Thus when lightning moves among dark clouds, the speed of its sinuous flight makes its whole course resemble a luminous snake. This is because the organ of perception acts more rapidly than the judgment." [During a moment of silence, the kiss is fully transformed into a cinematic kiss; the flicker disappears and continuity is dominant.] Through the persistence of vision, human perception is able to bridge the darkness, which always alternates, coequally with the light, on every motion picture screen.

At this point the shot fades out and is replaced by a shot of Muybridge walking across the screen repeatedly (reconstituted from his own sequential photographs) as the camera pulls back (Figures 6.14 and 6.15). The film ends with the observation in voice-over that "Muybridge transformed photography from the Zenonian reverie on movement it had been into the modern instrument that recovers the unity of human motion, the motion by which Zeno's paradoxes are refuted in a single step."

It matters little, I think, that persistence of vision is no longer considered to be an adequate explanation for the perception of movement in the cinema.⁴⁶ What matters for Andersen is that in the cinema darkness has been



6.14 Eadweard Muybridge,
Zoopraxographer.



6.15 Eadweard Muybridge,
Zoopraxographer.

banished and immobility has been successfully translated into mobility. Zeno has been refuted by cinematic technology. But what is most significant is Andersen's insistence on articulating this through the image of the human contact of a kiss. At the moment that the kiss becomes fully visible and continuous, the cinema begins. Yet this kiss is between two women; it is not the heterosexual kiss of Griffith, Porter, or the mainstream Hollywood cinema. Unlike Linda Williams, who sees Muybridge's work as the reinscription of scenarios that regulate the terms of a very conventional understanding of sexual difference,⁴⁷ Andersen emphasizes Muybridge's violation of Victorian norms and understandings of sexuality. Victorian society was, in Henry James's account cited by the film, a society whose "greatest triumph" was the "suppression of sex." Many of Muybridge's female models were prostitutes (as was common for artists of the period working with nudes). They have close-cropped hair and smoke cigarettes languorously, and Muybridge exploits the conventional association of prostitutes with lesbianism in scenarios in which two women walk together, arms around each other, or dance, or

kiss. But the scandal or radical subversiveness of Muybridge's work, for Andersen, is not, ultimately, lesbianism, but nakedness. According to the film's voice-over, Muybridge's work was "necessarily subversive of the taboo against realistic representation of nudity." For his society, materiality was equivalent to "evil," and nudity was therefore acceptable only if it was *de-materialized* through "lyricism, idealization, and good taste." There is a certain slippage here between nakedness, materiality, and, ultimately, radical materialism (the film begins with a quotation from Mao Zedong to the effect that "Man's knowledge of matter is knowledge of its forms of motion"). For Andersen, Muybridge's insistence upon nakedness is of the same order as the filmmaker's concern with the materiality of his or her own medium, which, in its turn, is compatible with the radical subversiveness associated with a Marxist materialism. Lesbianism would in this context be something of an afterthought, a further confirmation of Muybridge's assault on orthodoxy.

Nevertheless, in remaining faithful to Muybridge, in scrupulously miming his pseudoscientific grid, the gestures of the performers, and the fact that it is two women who kiss, Andersen's film makes visible an enabling mechanism of mainstream cinema's attempt to "bridge the darkness"—its insistent normalization of heterosexuality, a fact that strikes the film theorist with the force of an obviousness that makes it appear to resist analysis, to require no further dissection of its logic. Yet cinematic heterosexuality, at a very early stage, becomes fully imbricated in an attempt to grapple with the paradoxes of movement and temporality in the medium. Andersen's film calls attention to the absence of a lesbian kiss in most of the cinema following Muybridge, indeed, its contradiction or suppression through the constant reiteration of the heterosexual kiss as closure. In the classical cinema, it is the heterosexual kiss that "bridges the darkness."

A paradox is a statement that is contrary to received opinion or common sense but may actually be true. The etymology of the word traces it to the Greek *para* (beyond) and *doxon* (opinion). For Zeno, the kiss never really takes place; its suggestion of union is only illusory, because movement can never be born from a series of immobilities. Infinitely halving the distance between them, the women never reach each other. In the classical cinema, the kiss is consistently returned to as the quite particular disproof of the paradox of Zeno's philosophy, as the guarantee of the cinema's ability to

"bridge the darkness." The kiss here is the reassertion of the reigning doxa of family, race, nation, buttressed by a formal drive toward continuity.

Modernity has been delineated as the establishment of a social order based on mobility, flux, exchange. Perhaps this is why Walter Benjamin read Baudelaire's sonnet "A une passante" as exemplary of both the thematics of the crowd and the ephemeral temporality of the modern city. In it, a solitary woman emerges from the crowd, captures the gaze of the urban poet, and then disappears ("A lightning flash . . . then night"). The sonnet ends with the line "O you I would have loved (o you who knew it too!) [*O toi que j'eusse aimée, ô toi qui le savais!*]." Benjamin refers to this as "love—not at first sight, but at last sight."⁴⁸ Both the fascinations and the anxieties of modernity are figured as the failure of heterosexual contact, and all of this takes place in an instant. Benjamin derives his by now very familiar theory of shock in the course of his investigation of Baudelaire's poetry. And in this same essay he makes the following remark about photography: "The camera imparts to the instant an as it were posthumous shock."⁴⁹ As Samuel Weber has pointed out, the word translated as "instant" here is *Augenblick*, which is, literally, "eye-look," and has been translated as "in the blink of an eye."⁵⁰ In the crowded and "deafening" street, the urban poet loses his potential love "in the blink of an eye." Instantaneous photography, the basis of cinema, gives the instant a posthumous shock—it *takes* a picture, inscribes a moment (in the case of the cinema, eighteen to twenty-four per second). And the mechanism of the shutter in the camera (itself understood as a prosthetic extension of vision, as a figuring of the human eye) has been envisaged as "the blink of an eye." But the blink of an eye encompasses both light and darkness, vision and invisibility. The cinematic representation and celebration of mobility are founded on a basic stillness or immobility subtended by darkness. Its narrative denial of this is buttressed by the articulation of a series of imbricated obviousnesses—movement, instantaneity, heterosexuality, and visibility. The "flicker film" of the 1960s attempted to shatter this illusion and its allied comfort by assaulting the spectator at the level of perception. But many flicker films were contentless and failed to engage with the thematic matrix historically interpenetrating the illusion. Both modernity and the cinema have a stake in refuting Zeno, in affirming the reality, indeed the allure, of a mobility that is, in film, quite simply not there.